

Abode Flex 2.0 — API endpoint requirements

Every endpoint the Flex site and the admin need for Flex 2.0, Streaks & Points and Autopay: what exists, what must change, what is new, where each is used and why. **Admin endpoints now carry request and response shapes.**

Date: 6 Oct 2026 (status updated 8 Oct) · Revision of the 5 Oct 2026 document · Basis: Flex 2.0 PRD v0.1, Recorded Decisions (6 Oct 2026), Flex 2.0 admin mockups · Scanned: abode-v2, abode-fe-admin, abode-be-v2 at origin/staging (live OpenAPI) · Audience: backend, frontend and admin teams

How to read this. **EXISTS** verified as a route in the backend. **EXISTS · CHANGE** exists but must change for Flex 2.0. **NEW** does not exist; names and shapes are proposals for the backend team to confirm. **MISSING ON BE** the frontend already calls it but the backend has no such route. **REMOVED** was in the 5 Oct document and is no longer needed.

Scope rulings given by product (updated 6 Oct). The plan-terms snapshot for support and autopay management are **not admin screens**; the backend handles them. The admin app gets: the Offers tab pricing column, the Pricing editor, Pricing history, and **(D22) a Streaks & Points card in the user-details Summary with a streak-adjustment endpoint**. Emails are sent by the backend; the app saves no email preferences. D05 (new quote-change / pending-purchase expiry workflow) is deferred: keep the pricing snapshot and existing checkout handling.

What the 6 Oct decisions changed in this document. **D01** base tenor fixed at 36 months: no max-tenor field anywhere. **D02** admin enters only the base price; first and monthly payment are calculated (base ÷ 36). **D03** rounding settled: regular payments round down to 2 dp, final payment clears the residual. **Q9** no Finance approval reference: publishing uses the existing asset-pricing edit permission, enforced on the backend. **Q8** all Flex assets switch at launch: the rollout route and the features route are REMOVED. **Q11/D08** first installment is the minimum purchase payment, smaller later installments are accepted. **D04** no shorter-than-12-month Flex plan. **Q12** manual plans get an Enable-streaks flag. **Q13** property suggestions go only to customers with no active plan. Changed rows are tagged in their notes.

Status, 8 Oct 2026: implemented. The backend built the admin routes in this document on abode-be-v2 staging (PR #95) and they are live on the staging API. **Where the shipped behaviour differs from the text below, the backend wins:** (1) pricing writes use manage_assets and reads view_assets; (2) a draft is refused on a tenor_list size (409 PRICING_MODE_CONFLICT), and drafts are marked deferred by product; (3) streak adjust needs the new adjust_streak permission, takes a 10–500 character reason and an optional points_change (±100,000, never below a zero balance, else 409 POINTS_BELOW_ZERO); points are untouched without it; (4) there is no NO_ACTIVE_STREAK error, and the summary reports **0** for both streaks when no plan is streak-enabled; (5) enable_streaks is accepted on every manual plan type, not only Flex; (6) the pricing GET's legacy block adds tenor_36_unavailable_reason; (7) errors carry the code in code plus top-level errors[] / live_version; (8) the discounted total is rounded half up per unit, as assumed in section 3.4.

1. Summary

Area	EXISTS	EXISTS · CHANGE	NEW	MISSING ON BE	REMOVED
Customer app (Flex)	5	11	9	0	1
Admin app	1	9	8	1	1

Plus the backend-only work (section 4) and the email triggers (section 5). Each row is counted once; a few rows group routes that share a purpose. Admin rows are the endpoint blocks in section 3, each with request and response shapes.

2. Customer app — Flex site

Shapes for customer routes are unchanged from the 5 Oct document except where tagged **6 Oct**; request/response structures for these routes are not part of this revision.

2.1 Existing endpoints (reused or changed)

GET	/assets?offer_type=flex	EXISTS · CHANGE	Flex catalogue cards, landing	The card's "From ₦..." figure is today the minimum initial payment across stored plans. Sizes on a base plan must feed it from the base plan (the 36-month monthly payment).
GET	/assets/:id	EXISTS · CHANGE	Flex detail page, configure card	For a base-plan size also return base tenor (36), base price, checkpoints (24 and 12) and the live pricing version, so the page shows the standard 36-month plan before the budget field. Legacy sizes keep plans[]. 6 Oct : no max tenor. Same size fields as the admin tree (section 3.1).
GET	/assets/affordable	EXISTS · CHANGE	Affordability page	6 Oct (Q6) : must use Flex 2.0 pricing logic so the same estate, size, units and budget yield the same terms in discovery and purchase.
GET	/assets/trending	EXISTS	Explore, Home; "What's next" on a completed plan	6 Oct (D20) : use the existing endpoint and ranking; no new 30-day ranking. Owned estates and unavailable offers are excluded by the caller (e.g. the suggestions email job); take the highest-ranked eligible result.
GET	/acquisitions/my-plans	EXISTS · CHANGE	My assets list, Home, Wallet autopay plan picker	Add per-plan: pricing summary, covered_through, next obligation, autopay state, streak_enabled. Without these the Wallet picker needs one call per plan.
GET	/acquisitions/my-plans/:id	EXISTS · CHANGE	My plan page (coverage, timeline)	Add the saved pricing snapshot, per-month coverage including prepaid months, and autopay state. Timeline fields already exist (allocation_status, plan_completed_at, deed_sent_at).
GET	/acquisitions/flex/pending	EXISTS	Pending-transfer state after paying	Shows submitted but unconfirmed purchases. Must expose the saved quote so a pending purchase keeps its terms.
POST	/acquisitions/flex/paystack/initiate	EXISTS · CHANGE	Buy drawer (card)	Today takes tenor_months and looks up a stored plan. Must accept a server-issued quote reference and store the snapshot on the pending transaction. 6 Oct (Q11) : the first installment is the <i>minimum</i> payment; the customer may pay more, up to the agreed total, and the excess covers future installments without repricing. 6 Oct (D05) : no new QUOTE_CHANGED / expiry flow; keep existing checkout handling.
POST	/acquisitions/flex/transfer/submit	EXISTS · CHANGE	Buy drawer (bank transfer)	Same change as above. Approval later must use the submitted snapshot, not the current price.
POST	/acquisitions/flex/wallet/pay	EXISTS · CHANGE	Buy drawer (wallet)	Same change as above. A comment in the Flex frontend schema says no wallet purchase exists; that is stale — the endpoint exists and the hook uses it.
POST	/acquisitions/flex/paystack/recurring	EXISTS · CHANGE	Pay installment drawer, "Pay installment" buttons	6 Oct (D08, D11) : smaller payments are accepted and aggregate toward the monthly requirement; payments apply to the oldest unpaid installment first and must keep installments and penalties distinct. Must not collide with an in-flight autopay charge (extend the existing pending-recurring check).

POST	/acquisitions/flex/transfer/recurring · /wallet/recurring	EXISTS · CHANGE	Pay installment drawer	6 Oct: same partial-payment and oldest-first rules. A pending transfer pauses autopay for that plan (D18); approval uses the verified bank-payment date, not the approval date (D09).
GET	/acquisitions/flex/plans · /plans/:id · /plans/:id/documents(/:kind)	EXISTS	Owned plan and documents	Unchanged. Documents (contract of sale, certificate) are separate from payment completion.
POST	/payments/verify · /payments/paystack/verify	EXISTS	Return from Paystack	Unchanged.
GET	/wallet · /wallet/transactions	EXISTS · CHANGE	Wallet page (balance, history)	The Wallet page hosts “Set up autopay”. Transactions should show autopay charges; check whether a new transaction kind is needed. Wallet credits alone never earn points (Q5).
GET	/users/me/recent-activity	EXISTS	Home activity feed	Optional: show autopay results and points awards.

2.2 New endpoints

POST	/assets/:assetId/flex/quote	NEW	Flex detail — standard plan and budget calculator	Server-side pricing authority. Body: size_sqm, units, optional monthly_budget. Returns the plan that fits (months, discount, total, first/monthly/final payment), or too-low (with the minimum payment). 6 Oct (D04): at a budget equal to the 12-month installment return the 12-month plan; above it recommend the estate's actual Full Ownership offer while retaining the 12-month Flex option; if there is no matching FO offer keep the 12-month plan. Never invent a shorter plan or an FO price. Returns a signed quote reference bound to pricing version, size, units and duration.
GET	/users/me/streak	NEW	Streak & points page	Current and best streak, points balance, this month's qualification, per-plan coverage (what is covered, what is still needed) and upcoming covered months, shown separately from earned points (D12, Q2). One call, account-wide.
GET	/users/me/points/ledger	NEW	Points history	Append-only entries (month, points, reason, source). Adjustments and D13 reversals appear as separate entries. Paged.
POST	/autopay/authorization/initiate	NEW	Wallet → “Set up autopay” → authorize	Starts card authorization with Paystack and returns the redirect URL. Enables no plan. 6 Oct (D17): free setup where supported, or an existing reusable authorization with consent; otherwise a disclosed ₦50 charge that is refunded. Capability for Abode's account is still to be verified (Q10). Setup and refund never earn points.
POST	/autopay/authorization/verify	NEW	Return from authorization	Confirms the result and stores the reusable authorization. Drives the “setup failed / not finished” state. The webhook can also complete it.
GET	/autopay	NEW	Wallet autopay section	Authorization summary (brand, last four digits, status — never the token) plus every owned plan with enabled flag, next charge date, expected amount and last result.
PUT	/autopay/plans/:planId	NEW	Per-plan switch	Body {enabled}. Explicit consent per plan. Refused for suspended, completed or cancelled plans. A new purchase is never enabled automatically. 6 Oct (D19): for Full Ownership one property-level switch covers land installments and then document installments when their phase starts.

DELETE	/autopay/authorization	NEW	“Change card” / remove	Removes the authorization and stops every plan. 6 Oct (D18) : stop future charges and retries immediately; a charge already submitted may still resolve — tell the customer.
GET	/autopay/attempts	NEW	Recent autopay table	History of paid, failed, skipped (already covered) and confirming attempts for the customer.
GET	/users/me/features	REMOVED	—	Q8: all Flex assets follow the new flow at launch; no pilot cohort gating is needed.

3. Admin app

3.0 Conventions used by every shape below

Envelope	{ success, data, message, meta }. Shapes below show the contents of data. Lists are data: [...] with meta: { page, limit, total, totalPages }.
Money	Naira as a JSON number with at most 2 decimal places (the backend's existing IsNaira convention). Never strings, never kobo integers.
Percent	JSON number, e.g. 5 means 5%. Admin input has at most 2 decimal places; calculated discounts are returned to 4.
Dates	ISO-8601 UTC strings. Calendar months are "YYYY-MM" in Africa/Lagos.
Unknown fields	The backend rejects unknown body fields with 400, so the admin sends exactly the fields shown in each Request.
Permission	Pricing writes (PUT/DELETE draft, publish, convert-legacy) require the existing asset-pricing edit permission , enforced on the backend (Q9). The admin app gates the editor on manage_assets (view needs view_assets) — please confirm that is the right existing permission or give us its exact name. The streak card's Adjust button is gated on edit_user as a stand-in until you name the right one. Reads need asset-view. There is no Finance approval field .
Pricing routes	All new pricing routes sit under /admin/assets/:assetId/offers/flex/sizes/:sizeId/pricing, written ".../pricing".
Errors	Existing shape { statusCode, code, message[], path }. Codes proposed for this work are listed in 3.3.

3.1 Existing endpoints (reused, changed or missing)

GET `/admin/assets · /admin/assets/:id` **EXISTS · CHANGE** Admin: asset list and Offers tab

The Offers tab needs each Flex size's pricing mode and live version for the Pricing and Price version columns. The **list** response only needs pricing_mode and pricing.live_version on each size; the **detail** response returns the full object. Mockup mapping: "Base plan · ₦1,800,000 · 36 mo · 5% @ 24 · 15% @ 12" is built from pricing.base_price_per_unit and checkpoints; the "Pricing v2" badge from live_version.

Request GET /admin/assets/:id (no body) pricing_mode drives the Offers tab: base_plan -> one-line summary + "Edit pricing" tenor_list -> today's plans table + "Convert to base plan" unpriced -> new Flex size, shown as "Not priced yet"; must not be sellable until its first publish	Response (size object) // each size in offers[].sizes[] (additive; FO / commercial unchanged) { "_id": "665f01...", "size_sqm": 300, "units_available": 26, "is_active": true, "pricing_mode": "base_plan", // "base_plan" "tenor_list" "unpriced" "pricing": { // null unless pricing_mode = "base_plan" "live_version": 2, "base_tenor_months": 36, // always 36 in this release "base_price_per_unit": 3600000, "first_payment": 100000, // calculated; equals monthly_payment "monthly_payment": 100000, // months 1-35 "final_payment": 100000, // month 36, clears the residual "checkpoints": [{ "months": 24, "discount_pct": 5 }, { "months": 12, "discount_pct": 15 }], "published_at": "2026-10-05T09:12:00Z", "published_by": { "id": "u_1", "name": "A. Bello" }, "has_draft": false }, "plans": [] // [] for base_plan and unpriced; tenor plans for tenor_list }
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POST `/admin/assets/:assetId/offers/:offerType/sizes` **EXISTS · CHANGE** Admin: Add size

Today AddSizeDto requires plans, so the admin seeds a ₦1 placeholder plan. For a Flex offer, plans must become optional: a size created without plans is created **unpriced**. A Flex size sent *with* plans still works and becomes a tenor_list size. FO and commercial rules are unchanged.

Request { "size_sqm": 300, "units_available": 26 } // no "plans" for the new Flex flow	Response { "_id": "665f02...", "size_sqm": 300, "units_available": 26, "is_active": true, "pricing_mode": "unpriced", "pricing": null, "plans": [] }
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POST `/admin/assets/:assetId/offers` **EXISTS · CHANGE** Admin: Add offer (Flex)

Adding a Flex offer sends its first size inside the body (**OfferInputDto.sizes[]**), and every nested size is a `SizeInputDto`, so the same relaxation as Add size applies: for `offer_type: "flex"`, `sizes[].plans` becomes optional and a size without plans is created **unpriced**. Everything else about adding an offer is unchanged.

Request <pre>{ "offer_type": "flex", "is_active": true, "allocation_qualification_pct": 30, "sizes": [{ "size_sqm": 300, "units_available": 26 } // no "plans"] }</pre>	Response <pre>201 the created offer, each size with "pricing_mode": "unpriced", "pricing": null, "plans": []</pre>
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PATCH / DELETE `/admin/assets/:assetId/offers/:offerType/sizes/:sizeId` **EXISTS · CHANGE** Admin: Edit / delete size

Units and active flag keep working in every pricing mode (no new fields). A PATCH that full-replaces `plans[]` on a `base_plan` size is refused. Deleting a size that has purchases keeps its existing guard.

Request PATCH body (unchanged) <pre>{ "units_available": 30, "is_active": true }</pre> PATCH on <code>base_plan</code> size with <code>"plans": [...]</code>	Response <pre>200 updated size (shape above) 409 { "code": "PRICING_MODE_BASE_PLAN", "message": ["This size is priced by a base plan. Edit its pricing instead."] }</pre>
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POST / PATCH / DELETE `/admin/assets/:assetId/offers/:offerType/sizes/:sizeId/plans(/:tenor)` **EXISTS · CHANGE** Admin: Add / edit plan (tenor list)

Kept for legacy `tenor_list` sizes only, so there is one source of truth. Bodies are unchanged (`PlanInputDto`: `tenor_months`, `land_price`, `initial_payment`, `monthly_installment`, `is_promo`, `is_active`). On a `base_plan` or `unpriced` size return the 409 below.

Request existing <code>PlanInputDto</code> – unchanged	Response <pre>409 { "statusCode": 409, "code": "PRICING_MODE_BASE_PLAN", "message": ["Plans cannot be edited on a base-plan size."] }</pre>
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GET	/admin/assets/:assetId/offers/history	MISSING ON BE	Admin: Offers → History sheet
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The admin already calls this (paged, newest first); staging has no such route. Existing actions are kept (add-offer, update-offer, add-size, update-size, delete-size, add-plan, update-plan, delete-plan); **two are added: publish-pricing and convert-pricing**. The mockup shows each pricing entry with its checkpoints and base price in the summary, a purchase count ("3 purchases so far"), a "now superseded" marker, and an alert about buyers — including transfers awaiting approval — who keep their saved terms; those come from purchase_count, superseded and pending_transfer_count. The summary stays server-written; the stored snapshot is read from the versions routes below.

Request GET .../offers/history?page=1&limit=50	Response <pre>{ "data": [{ "version": 17, "action": "publish-pricing", "summary": "300 sqm · Pricing v2 – 5% @ 24, 15% @ 12 · base ₦3,600,000", "changed_by": "A. Bello", "changed_at": "2026-10-05T09:12:00Z", "size_id": "665f01...", "pricing_version": 2, "purchase_count": 3, "pending_transfer_count": 0, "superseded": false }, { "version": 14, "action": "update-plan", "summary": "500 sqm · 24 months: monthly ₦260,000 → ₦265,000", "changed_by": "F. Ade", "changed_at": "2026-09-02T11:02:00Z" }], "meta": { "page": 1, "limit": 50, "total": 5, "totalPages": 1 } }</pre>
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GET	/admin/payment-plans · /summary · /export	EXISTS	Admin: Payment Plans page
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No new admin screen. Optional: add pricing_version and streak_enabled to each row.

GET `/admin/flex/purchase/plans · /plans/:planId` **EXISTS · CHANGE** Admin: plan detail (via users and payment plans)

The response carries the saved pricing snapshot read-only. Per the scope ruling there is no admin screen for it; support reads it from the API. For manual plans source is `admin_created` and the snapshot holds the terms used at creation (Q12).

Request (no body)	Response (plan) <pre>{ ...existing plan fields, "streak_enabled": true, "pricing_snapshot": { "source": "quote", // "quote" "admin_created" "legacy_plan" "pricing_version": 2, // null for admin_created, legacy_plan "base_price_per_unit": 3600000, "checkpoints": [{ "months": 24, "discount_pct": 5 }, { "months": 12, "discount_pct": 15 }], "entered_monthly_budget": 350000, // null if none entered "selected_tenor_months": 18, "applied_discount_pct": 10, "units": 2, "total": 6480000, "first_payment": 360000, "monthly_payment": 360000, "final_payment": 360000, "method_version": "piecewise-linear-v1" } }</pre>
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POST `/admin/users/:id/assets/flex` **EXISTS · CHANGE** Admin: admin-created plan

A manually created plan has no customer quote, so its snapshot source is `admin_created`. **6 Oct (Q12)**: add an optional “Enable streaks” flag, on by default; off means the plan neither earns nor blocks the account-wide streak. Autopay is never enabled by creation. All existing `AdminCreateFlexPlanDto` fields are unchanged.

Request <pre>{ ...all existing AdminCreateFlexPlanDto fields..., "enable_streaks": true // NEW, optional, default true }</pre>	Response <pre>{ ...created plan, "streak_enabled": true, "pricing_snapshot": { "source": "admin_created", "pricing_version": null, "selected_tenor_months": 24, "total": 2400000, "first_payment": 100000, "monthly_payment": 100000 } }</pre>
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POST	<code>/admin/acquisitions/transactions/:txId/approve</code> · <code>/decline</code>	EXISTS · CHANGE	Admin: transfer approval
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No change to the admin request or response. Behaviour: approval must create the plan from the *submitted* snapshot even when a newer pricing version is live, and uses the verified bank-payment date, not the approval date, for streak purposes (D09). A transfer awaiting review pauses autopay for that plan; a decline must notify the customer before collection resumes (D18).

PATCH / POST	<code>/admin/users/:id/assets/:planId/next-payment-date</code> · <code>balance/adjust</code> · <code>spec</code> · <code>suspend</code> · <code>unsuspend</code> · <code>close</code>	EXISTS · CHANGE	Admin: plan mutations
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No change to requests or responses. Behaviour: each must recompute coverage, re-evaluate the affected month's streak record, and pause or stop autopay. A suspended streak-enabled plan with an outstanding balance blocks account-wide qualification until reinstated (D11). A closed or soft-deleted plan stays in assessments until closure is finalised and nothing collectible remains. Reward removals go through the points ledger as visible entries (D13), never a silent edit.

3.2 New endpoints

GET

.../pricing

NEW

Admin: Pricing editor (open sheet)

Returns what the editor sheet needs in one call: limits, the live version, the current draft, and for tenor-list sizes the legacy data used to pre-fill “Convert to base plan”. Mockup mapping: the “Base tenor” field is read-only from limits.base_tenor_months (D01); there is no Maximum-tenor field. The sheet's “Draft v3 · based on v2” badge is draft.next_version and draft.based_on_version. **legacy.tenor_36_land_price** is the existing 36-month plan price per unit; when it is null or inconsistent the editor shows a blank base price and says why — a price is never inferred (Q7).

Request

(no body, no query)

Response

```
{
  "size_id": "665f01...",
  "size_sqm": 300,
  "pricing_mode": "base_plan",
  "limits": {
    "base_tenor_months": 36,
    "min_tenor_months": 12,
    "checkpoint_months": [24, 12]
  },
  "live": {
    "version": 2,
    "base_price_per_unit": 3600000,
    "first_payment": 100000,
    "monthly_payment": 100000,
    "final_payment": 100000,
    "checkpoints": [
      { "months": 24, "discount_pct": 5 },
      { "months": 12, "discount_pct": 15 }
    ],
    "method_version": "piecewise-linear-v1",
    "published_by": { "id": "u_1", "name": "A. Bello" },
    "published_at": "2026-10-05T09:12:00Z",
    "purchase_count": 3
  },
  "draft": null,
  "legacy": null
}
// or the draft object (see PUT draft)
// only when pricing_mode = "tenor_list":
// { "tenor_36_land_price": 3600000 | null,
//   "active_tenors": [12, 24, 36] }
```

POST

.../pricing/preview

NEW

Admin: Pricing editor (preview table)

Server computes every whole month from 12 to 36: discount, total, regular and final payment, and the largest single payment. The page also calculates instantly in the browser; this proves the server agrees with it. Mockup mapping: the table columns Months / Discount / Total / Payment / Final payment / “Needs ≥ / mo” are months, discount_pct, total, regular_payment, final_payment, largest_payment; is_base and is_checkpoint drive the “base” and “checkpoint” tags and shading. “Needs ≥ / mo” is largest_payment, so a customer’s entered budget always covers the plan shown. Preview never throws on bad numbers; it returns valid:false and the same error list publish would return.

Request <pre>{ "base_price_per_unit": 3600000, "checkpoints": [{ "months": 24, "discount_pct": 5 }, { "months": 12, "discount_pct": 15 }] } // same body as draft and publish. // Price is per unit. Month values are fixed // by the server (24 and 12); only the // discounts are editable.</pre>	Response <pre>{ "valid": true, "errors": [], "first_payment": 100000, "monthly_payment": 100000, "final_payment": 100000, "rows": [{ "months": 36, "is_base": true, "is_checkpoint": false, "discount_pct": 0, "total": 3600000, "regular_payment": 100000, "final_payment": 100000, "largest_payment": 100000 }, { "months": 35, "is_base": false, "is_checkpoint": false, "discount_pct": 0.4167, "total": 3585000, "regular_payment": 102428.57, "final_payment": 102428.62, "largest_payment": 102428.62 }, { "months": 24, "is_checkpoint": true, "discount_pct": 5, ... }, ...one row per whole month, 36 down to 12...] } // invalid numbers: 200 with "valid": false, "errors": // [{ "field", "code", "message" }] and "rows": []</pre>
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PUT / DELETE

.../pricing/draft

NEW

Admin: Pricing editor → Save draft

Saves the base price and the two discounts without affecting customers. One shared draft per size; PUT replaces it. **Also allowed on a tenor_list size**, so a conversion can be prepared ahead of launch (Q8) and published later with convert-legacy. A draft is checked for shape only (numbers present, not negative); the full rules run on publish. DELETE discards it.

Request <pre>{ "base_price_per_unit": 3600000, "checkpoints": [{ "months": 24, "discount_pct": 5 }, { "months": 12, "discount_pct": 15 }] } // same body as draft and publish. // Price is per unit. Month values are fixed // by the server (24 and 12); only the // discounts are editable.</pre>	Response <pre>{ "based_on_version": 2, // null if unpriced "next_version": 3, // what Publish will create "base_price_per_unit": 3700000, "checkpoints": [{ "months": 24, "discount_pct": 5 }, { "months": 12, "discount_pct": 15 }], "saved_by": { "id": "u_1", "name": "A. Bello" }, "saved_at": "2026-10-06T08:00:00Z" } DELETE .../pricing/draft -> 204 (discard the draft)</pre>
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POST

.../pricing/publish

NEW

Admin: Pricing editor → Publish

Validates, then creates a new **immutable** version and makes it live. Existing and pending purchases are untouched. `expected_live_version` guards two admins publishing at once. Validation (every problem is returned, not just the first): base price > 0 with at most 2 dp; exactly two checkpoints at months 24 and 12; each discount ≥ 0 and < 100 with at most 2 dp; and a shorter plan never costs more, i.e. $0 \leq d_{24} \leq d_{12}$. There is no tenor, reconciliation or Finance-reference rule (D01, D02, Q9). Mockup mapping: the red messages (“Two checkpoints are both at...”, “The 24-month discount is larger than the 12-month discount...”) are the server’s error list; the Publish button stays disabled client-side while the same rules fail.

Request

```
{
  "base_price_per_unit": 3600000,
  "checkpoints": [
    { "months": 24, "discount_pct": 5 },
    { "months": 12, "discount_pct": 15 }
  ],
  "expected_live_version": 2      // null for the first publish
}
// no approval reference, no tenor, no first or
// monthly payment: those are not inputs any more.
```

Response

```
201
{
  "version": 3,
  "based_on_version": 2,
  "size_id": "665f01...",
  "base_tenor_months": 36,
  "base_price_per_unit": 3600000,
  "first_payment": 100000,
  "monthly_payment": 100000,
  "final_payment": 100000,
  "checkpoints": [ ... ],
  "method_version": "piecewise-linear-v1",
  "published_by": { "id": "u_1", "name": "A. Bello" },
  "published_at": "2026-10-06T09:00:00Z",
  "purchase_count": 0,
  "status": "live"
}
// 400 PRICING_VALIDATION_FAILED, 409 PRICING_VERSION_CONFLICT
// (see 3.3). The previous version becomes "superseded".
```

GET

.../pricing/versions · .../pricing/versions/:version

NEW

Admin: Pricing history

Version list with purchase counts, and a stored snapshot per version so any historical quote can be reproduced. Backs the “Pricing history” detail behind the History sheet (the sheet’s list itself comes from offers/history above).

Request (no body)	Response GET .../pricing/versions?page=1&limit=20 { "data": [{ "version": 2, "status": "live", "base_price_per_unit": 3600000, "checkpoints": [...], "published_by": { "id": "u_1", "name": "A. Bello" }, "published_at": "2026-10-05T09:12:00Z", "purchase_count": 3, "pending_transfer_count": 0 }, { "version": 1, "status": "superseded", ... "purchase_count": 11, "pending_transfer_count": 2 }], "meta": { "page": 1, "limit": 20, "total": 2, "totalPages": 1 } } GET .../pricing/versions/2 the publish response shape (full snapshot) plus "status", "purchase_count", "pending_transfer_count", so any historical quote can be reproduced. 404 PRICING_VERSION_NOT_FOUND for an unknown version.
---	--

POST

.../pricing/convert-legacy

NEW

Admin: Offers tab → “Convert to base plan”

Moves a hand-entered tenor-list size onto a base plan. It is atomic: validation, version 1, the mode switch and the history entry (action convert-pricing) all succeed or none do. Existing buyers keep their original plans and terms; the old tenor plans stay on the backend for those plans, are no longer returned in plans[], and can no longer be sold. Every Flex size must be prepared before launch (Q8), so this is the route the pre-launch data task uses; sizes with missing or inconsistent 36-month data must be resolved explicitly, not guessed (Q7).

Request { "base_price_per_unit": 3600000, "checkpoints": [{ "months": 24, "discount_pct": 5 }, { "months": 12, "discount_pct": 15 }] } // expected_live_version is not accepted: // a tenor_list size has no live version.	Response 201 { "version": { ...publish response shape, version = 1 }, "size": { "_id": "665f01...", "pricing_mode": "base_plan", "plans": [] } } 409 PRICING_MODE_CONFLICT size is already on a base plan 400 PRICING_VALIDATION_FAILED same rules as publish
--	--

GET `/admin/users/:id/streak` **NEW** Admin: user details → Summary → Streaks & Points card

6 Oct (D22). Data for the card that shows the active streak and points balance. Read-only; requires the user-view permission.

Request	Response
(no body)	<pre>{ "current_streak": 7, "best_streak": 9, "points_balance": 1300, "current_month": { "month": "2026-10", "qualified": true, "awarded_at": "2026-10-03T11:20:00Z" }, "covered_through": "2026-12", // last prepaid month; null if none "streak_enabled_plans": 2, "last_adjustment": { // null if never adjusted "at": "2026-09-30T10:00:00Z", "by": "F. Ade", "reason": "Support ticket 4411" } }</pre> // no streak-enabled plan: zeros, "current_month": null, // "streak_enabled_plans": 0 -> the card shows an empty // state, never a faked zero streak.

POST `/admin/users/:id/streak/adjust` **NEW** Admin: user details → Streaks & Points card → Adjust

6 Oct (D22). Authorised streak adjustment with a required reason and an append-only audit record (actor, user, before, after, reason, time) that also appears in Admin Logs. The decisions do **not** approve arbitrary points adjustment, adjustment limits, or an automatic points change when a streak is adjusted, so `points_balance` never changes here. System-led corrections for reversed payments stay governed by D13.

Request	Response
<pre>{ "current_streak": 5, // integer >= 0: the value the // streak should become "reason": "Bank transfer verified on time, approved late (ticket 4411)" // required, 20 to 2000 characters (the admin's // existing reason rule) }</pre>	<pre>200 { "before": { "current_streak": 3, "best_streak": 9, "points_balance": 1300 }, "after": { "current_streak": 5, "best_streak": 9, "points_balance": 1300 }, "audit_id": "adj_8f3c01", "adjusted_at": "2026-10-06T10:15:00Z", "adjusted_by": { "id": "u_1", "name": "A. Bello" } }</pre> 409 NO_CHANGE requested value equals the current streak 409 NO_ACTIVE_STREAK the customer has no streak-enabled plan 403 caller lacks the adjustment permission

PATCH `/admin/config/flex-2-rollout` **REMOVED** —

Q8: every Flex asset follows the new flow at launch and there is no per-estate pilot switch. Preparing every size's configuration before launch replaces the pilot gate.

3.3 Error codes proposed for the admin routes

HTTP	code	When
400	PRICING_VALIDATION_FAILED	Publish or convert-legacy failed validation. Body also carries errors: [{ field, code, message }] listing every problem. Rule codes: BASE_PRICE_INVALID, CHECKPOINTS_INVALID, DISCOUNT_OUT_OF_RANGE, DISCOUNT_ORDER_INVALID.
409	PRICING_VERSION_CONFLICT	expected_live_version is stale. Body includes live_version so the app can reload.
409	PRICING_MODE_CONFLICT	A pricing route on a non-Flex size, publish on a tenor_list size, or convert-legacy on a base_plan size.
409	PRICING_MODE_BASE_PLAN	Plan add / edit / delete, or full-replace of plans[], on a base-plan or unpriced size.
404	PRICING_VERSION_NOT_FOUND	Unknown version.
409	NO_CHANGE	Streak adjust to the value the streak already has (no audit record is written).
409	NO_ACTIVE_STREAK	Streak adjust for a customer with no streak-enabled plan: there is no active streak to change.
403	(existing permission error)	Caller lacks the pricing-edit or streak-adjustment permission.

3.4 Pricing engine the admin shapes assume

One 36-month base plan per Flex size, with admin-set discounts at exactly two checkpoints: 24 and 12 months. Any whole month from 12 to 36 is calculated by straight-line interpolation between the two nearest points of (36, 0%), (24, d24) and (12, d12), separately for each segment (D01–D03). Use decimal arithmetic, not floats.

```

discount(n) = d24 * (36 - n) / 12      for 24 <= n <= 36
discount(n) = d24 + (d12 - d24) * (24 - n) / 12    for 12 <= n < 24
total(n)    = round_half_up_to_kobo( base_price * (1 - discount(n) / 100) )    per unit
regular(n)  = floor_to_2dp( total(n) / n )    months 1 .. n-1 (month 1 is the first payment)
final(n)    = total(n) - regular(n) * (n - 1)    month n, clears the exact residual
largest(n)  = max(regular(n), final(n))

```

Rounding the *discounted total* itself to the kobo is our assumption (D03 fixes only installment rounding) — please confirm, see section 8. Reference vectors (base ₦3,600,000 per unit, d24 = 5%, d12 = 15%):

Months	Discount	Total	Regular payment	Final payment
36	0.0000%	3,600,000.00	100,000.00	100,000.00
30	2.5000%	3,510,000.00	117,000.00	117,000.00
24	5.0000%	3,420,000.00	142,500.00	142,500.00
22	6.6667%	3,360,000.00	152,727.27	152,727.33
18	10.0000%	3,240,000.00	180,000.00	180,000.00
12	15.0000%	3,060,000.00	255,000.00	255,000.00

Also from the decisions record: ₦1,000,000 over 36 months at 0% gives 35 payments of ₦27,777.77 and a final payment of ₦27,778.05. For a 5% / 15% configuration, 30 months is 2.5% and 18 months is 10%, as in the table.

3.5 How each mockup element is fed

Mockup screen · element	Field / route
1 Offers · offer header “Flex · Active · Allocation qualification 30%”	offer.offer_type, offer.is_active, offer.allocation_qualification_pct (existing)
1 Offers · size, units available, status	size.size_sqm, size.units_available, size.is_active (existing)
1 Offers · Pricing cell (“Base plan” + ₺ · 36 mo · discounts)	size.pricing_mode = base_plan, pricing.base_price_per_unit, pricing.checkpoints
1 Offers · Pricing cell (“Tenor list · 3 hand-entered plans”)	size.pricing_mode = tenor_list, size.plans[].tenor_months
1 Offers · Price version badge (“Pricing v2” / —)	pricing.live_version / null
1 Offers · Edit pricing / Convert to base plan	GET .../pricing, then publish; or convert-legacy for tenor_list
1 Offers · + Add size	POST .../sizes with no plans (unpriced size)
2 Editor · size chips	offers → sizes[] already loaded; one GET .../pricing per size
2 Editor · “Draft v3 · based on v2”	draft.next_version, draft.based_on_version
2 Editor · Base tenor 36 (read-only)	limits.base_tenor_months. Maximum tenor field removed (D01)
2 Editor · Base price per unit	publish / draft / preview body: base_price_per_unit
2 Editor · First and monthly payment	Read-only, calculated (D02): preview first_payment, monthly_payment, final_payment. Reconciliation check removed
2 Editor · Discount checkpoints	checkpoints[]: months fixed at 24 and 12, discount_pct editable
2 Editor · Preview table	POST .../pricing/preview rows[]
2 Editor · validation list	preview.errors[] / publish 400 errors[]
2 Editor · Finance approval reference	Removed (Q9)
2 Editor · Save draft / Publish as v3	PUT .../pricing/draft / POST .../pricing/publish
2 Editor · “Load example” chips	Demo-only in the mockup; not built
3 History · entries, purchases, superseded, transfers note	GET .../offers/history (publish-pricing, convert-pricing, purchase_count, superseded, pending_transfer_count)
User details · Streaks & Points card (D22, not in the mockup)	GET /admin/users/:id/streak, POST /admin/users/:id/streak/adjust

4. Backend-only work (no endpoint called by the apps)

Pricing engine	Piecewise-linear interpolation at full precision, rounded to the kobo. 6 Oct (D03, settled) : regular payments round down to 2 dp, the final payment clears the residual; applies to the base plan and generated shorter plans. The backend already accepts two-decimal naira (IsNaira).
Quote signing	Sign and verify the quote reference; short life; bound to version, size, units and duration. 6 Oct (D05) : no new expiry workflow; keep the snapshot and existing checkout handling.
Snapshot writer	Inside the existing initial-purchase settlement transaction, so a plan and its snapshot are created together or not at all. Manual plans store source admin_created.
Coverage service	One definition of “covered”: confirmed payments applied to the saved schedule, including prepaid months. 6 Oct (D08, D10, D11) : smaller payments aggregate; oldest unpaid installment first; penalties do not raise the target; a final balance below the installment covers the last month; suspended plans with a balance block account qualification; FO document installments count only once their schedule starts. Used by the streak job, the autopay collector and the reminder emails.
Monthly streak evaluator	One qualification record per customer per month (unique key), Africa/Lagos. Awards 100 points once, as soon as all eligible plans are covered (D07); future prepaid months are awarded as they arrive, not early (Q2). Bank-transfer months use the verified payment date (D09). A missed month resets the streak; best streak and points remain (D12). Idempotent against repeated runs and callbacks.
Payment reversals	6 Oct (D13) : re-evaluate affected months; if qualification is lost, remove the month's 100 points via a visible adjustment, recalculate the streak and notify the customer; if other confirmed payments preserve coverage, keep the reward.
Launch backfill	6 Oct (D14) : include existing Flex and FO customers from the launch month onward; no rewards for earlier months; existing prepayments count for launch-month and future coverage.
Points ledger	Append-only. Corrections are new entries with actor, reason and before/after. Setup and refunded verification charges never earn points.
Streak adjustment	6 Oct (D22) : the admin adjust endpoint writes an audit record and a ledger-visible entry; it never changes points unless product approves that.
Autopay collector (daily)	Rechecks coverage and the collectible amount; skips covered months and plans with a pending or in-flight payment or a pending transfer; charges no more than the remaining balance; stops on completion; never initiates for suspended or cancelled plans. 6 Oct (D18) : retries at 24 h and 72 h after the <i>original</i> failed attempt, recheck coverage before each retry, stop after two retries, never retry an unresolved result. Charges the authorized card directly via Paystack, not through a Wallet top-up (D16).
Autopay reconciler	Resolves uncertain attempts before any further charge; protects against duplicate collection.
Webhook handling	Extend POST /webhooks/paystack (today it routes charge.success to flex, FO, marketplace and referral by reference) to autopay events, idempotent by reference.
Suggestions ranking	6 Oct (D20, Q13) : use the existing hottest-assets endpoint and ranking; exclude owned estates and unavailable offers; take the highest-ranked eligible result; send nothing if none exists. Audience: customers who completed payment and have no current active plan.
Pre-launch data task	6 Oct (Q7, Q8) : before switching, configure every existing Flex size (convert-legacy). Existing 36-month prices are the starting point, subject to validation; missing or inconsistent data is resolved explicitly, not inferred.

5. Emails (sent by the backend)

No endpoint is involved and the app saves no preferences. Timings are the recorded product decisions.

Email	When	Status	Notes
Payment received	As soon as the customer's payment is confirmed	Decided	Templates exist today (flexPaymentConfirmation, flexReceipt). Check they say "month n of N" and fire at settlement.
Autopay reminder	One notice 24 hours before the charge	Decided (Q1)	Combines the earlier "day or two" and "day before" notices; shows property, amount and date. Routine pre-payment reminders are suppressed for autopay plans (Q3).
Failed deduction	Immediately on a confirmed failure, and after each retry	Decided	Stop after the two retries and direct to manual payment or updating authorization. Do not duplicate failure messages.
Streak celebration	As soon as a payment qualifies the month	Decided	100 points and the correct streak count. Playful tone; payment and failure notices stay accurate and clear.
Prepaid month recognised	As each covered month arrives	Decided (Q2)	No wait until month-end and no extra payment.
Streak reminder	One week before the due date	Decided	Combine with nearby existing reminders (15, 10, 3 days, due-today); suppress for covered installments. Account-wide handling of multiple due dates is an implementation detail.
Streak lost (reversal)	When D13 removes a month's points	Decided	Notify the customer of the adjustment.
Property suggestions	Every two weeks	Decided (Q13)	Only customers who completed payment and have no active plan ; stop once they start a plan. Scheduler anchor (global batch vs 14 days from last send) not yet selected.
Final payment	With the final payment	Decided	Statement, congratulations and certificate emails exist today. Confirm there is no duplicate celebration; do not tell a completed customer they must buy again to keep their streak.

6. Data the backend must add

PaymentPlan (extend)	pricing_snapshot (source, version, checkpoints, method version, entered budget, selected tenor, applied discount, schedule), applied_discount_pct, quote reference, coverage fields, autopay_state, streak_enabled (default true)
FlexSize (extend)	pricing_mode (base_plan tenor_list unpriced); base plan (price per unit; base tenor is the constant 36); live version pointer; one draft. No max tenor.
FlexPricingVersion (new)	immutable version records: size, version, base price per unit, checkpoints, method version, published by and when, status. No approval reference (Q9).
StreakMonth (new)	unique (user, month): qualified flag, plans evaluated, coverage per plan, awarded at
PointsLedger (new)	append-only: user, month, points, kind (earn / adjustment / reversal), reason, source, actor
StreakAdjustmentAudit (new)	append-only: user, actor, before and after streak, reason, created at; surfaced in Admin Logs
AutopayAuthorization (new)	user, provider token (never exposed), brand, last four, status, created
AutopayPlanSetting (new)	user, plan, enabled, enabled at, last result
AutopayAttempt (new)	plan, obligation month, amount, reference, outcome (paid / failed / skipped / confirming), reason, retry count, original failed attempt reference

7. Findings from the code scan

- **Offer history route is missing.** The admin calls GET /admin/assets/:id/offers/history; staging has no such route. The existing log would also lack a snapshot to reproduce quotes — hence the pricing versions routes.
- **The History button is mock-only in the admin today.** It is shown only when mocks are on, and flips to real once the route exists.
- **Purchases are keyed to a stored tenor.** The purchase endpoints look up an active plan by tenor_months. A budget-generated 20-month plan has no stored plan to find.
- **The first payment is free-form today.** The caller sends any amount from the required initial up to the total. Flex 2.0 makes the first installment the minimum and lets the customer pay more (Q11).
- **No reusable card authorization is captured.** Paystack initialization requests no reusable card and the webhook only acts on charge.success for purchases. A wallet field paystack_authorization_code exists but holds only migrated v1 data.
- **Wallet top-up was removed.** Confirm which money still lands in the Wallet (commission, marketplace sales, refunds) so the “deposits don't count” rule is worded correctly.
- **Penalties and suspension already run.** flex-defaulting adds penalties from 2 months overdue and suspends at 4. Streak eligibility and autopay must respect D11.
- **Reminder emails already exist** at 15, 10, 3 days, due-today and overdue. New emails need the de-duplication rules in section 5.
- **Flex size creation needs a plan.** The admin seeds a ₦1 placeholder plan to create a size (OfferEditDialogs). A base-plan size must not need one.
- **Admin permissions are not enforced in the admin app** (documented in its CLAUDE.md). Publishing prices must be permission-gated on the backend with a named permission.
- **Stale comment in the Flex frontend.** flex.schema.ts says no wallet-funded purchase exists; the endpoint and hook do.

8. Open items

Product-level items the decisions record leaves open, then items the backend team needs to answer for the admin shapes.

#	Item	Working assumption in this document
1	Exact name of the existing asset-pricing edit permission (Q9), and which existing permission controls streak adjustment (D22).	Pricing writes use the pricing-edit permission; streak adjustment uses a named permission. No numeric adjustment limits.
2	Rounding of the discounted <i>total</i> to the kobo. D03 settles installment rounding only.	Round half up to the kobo, then floor regular payments to 2 dp.
3	Is plan land_price per unit? The pricing contract is per unit.	Per unit, matching how sizes are priced today.
4	Does adjusting a streak above best_streak raise best_streak? Does adjusting a streak change points (not approved).	best_streak = max(best, new current); points never change through this route.
5	Tenor plans retained after convert-legacy: referenced by existing purchases only?	Yes; hidden from plans[] and no longer sellable.
6	Existing Flex sizes with no 36-month plan or inconsistent plans (Q7).	Convert is blocked for the admin until a base price is entered; nothing is inferred.
7	Draft ownership: one shared draft per size, or one per admin.	One shared draft per size.
8	Q10 / D17: reusable-card and zero-charge setup support on Abode's Paystack account.	Technical verification outstanding; not an admin concern.

#	Item	Working assumption in this document
9	Recommendation scheduler anchor; clock time of awards and collection jobs; refund timing and fees for the ₦50 setup charge; goodwill-refund classification; months with no obligation between early land payoff and FO document start.	Left open by the decisions record; not assumed here.

9. Suggested build order

1. Pricing engine, versioned base plan, publish and preview (admin), plus Flex size creation without a plan — unblocks everything else.
2. Quote endpoint, then purchase endpoints accepting a quote and writing the snapshot (customer purchase). Update Affordability to the same calculation (Q6).
3. Pre-launch data task: convert every existing Flex size (Q7, Q8).
4. Coverage service, streak evaluator and points ledger, then the customer read endpoints and the admin streak summary and adjust endpoints.
5. Offer-history route, with pricing events.
6. Autopay: D16 and D18 are decided; D17 needs the Paystack capability check (Q10). Build authorization, per-plan enablement and the collector last.
7. Emails, in parallel with the above. (The rollout switch is no longer needed.)

Route names and bodies for NEW items are proposals. Existing routes were checked against the live staging OpenAPI on 6 Oct 2026. Request and response shapes for admin routes were derived from the Flex 2.0 admin mockups and the 6 Oct decisions record.